

**Applicability of conditional diffusion
model for climate statistical downscaling
in Canary Islands.**

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Resumen

Los modelos climáticos globales son esenciales para el estudio de los efectos resultantes del cambio climático a larga escala. Sin embargo, la baja resolución de estos modelos lleva a que no proporcionen una descripción representativa de estos efectos a escala regional, ni en terrenos con una orografía complicada. Es por esto que se introducen la regionalización climática dinámica y estadística. La regionalización dinámica está basada en los modelos climáticos regionales, que presentan grandes costes de computación, mientras que la regionalización estadística se reduce a estudiar las relaciones empírico-estadísticas de las variables termodinámicas, sin proporcionar ecuaciones físicas.

En este trabajo se aprovecha el auge de los algoritmos de aprendizaje automático y profundo que han surgido en los últimos años en el campo, para implementar un modelo de inteligencia artificial generativa denominado *Denoising Diffusion Probabilistic Model*, condicionado mediante variables sinópticas, y aplicado a la regionalización climática estadística en las Islas Canarias, concretamente, a las islas de Tenerife, Gran Canaria, y La Palma, regiones muy pequeñas en términos globales, y con una orografía compleja.

El modelo, con una estructura relativamente simple, es capaz de generar mapas de temperatura media diaria a alta resolución, mediante las condiciones sinópticas a escala global, a poca resolución. Se han calculado las métricas proporcionadas por la iniciativa europea VALUE, y se llega a la conclusión de que los mapas de temperatura media inferidos por el modelo presentan métricas satisfactorias cuando se comparan con las simulaciones de los modelos mesoescalares de WRF, consideradas pseudo-observaciones, lo que sugiere que el tipo de modelos planteados sean muy prometedores para realizar proyecciones climáticas.

Abstract

Global Climate Models (GCMs) are essential to study climate change effects at large scale. The coarse spatial resolution of GCMs leads to the fact that they do not provide a representative description of these effects on regional scales, or terrains with a complex orography. Thus, to solve this problem, statistical and dynamical climate downscaling are introduced. Dynamical downscaling is based on Regional Climate Models (RCMs), which present high computational costs, while the statistical downscaling models are reduced to study de empirical-statistical relations of the thermodynamic variables, without providing them with physical equations.

This work takes advantage of the rise of machine and deep learning algorithms, to implement a generative artificial intelligence model, *Denoising Diffusion Probabilistic Model* (DDPM), conditioned on large scale synoptic conditions, and applied to the statistical downscaling in Canary Islands, namely, the islands of Tenerife, Gran Canaria, and La Palma, small regions with a complex orography.

The model, with a relatively simple structure, is able to generate daily mean temperature maps at high resolution, via synoptic conditions at global scale and low resolution. The metrics provided by the European VALUE initiative have been calculated, and it is concluded that the mean temperature maps inferred by the model present satisfactory metrics when compared to WRF mesoscale model simulations, considered pseudo-observations, suggesting that the kind of models proposed are very promising for climate projections.

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I Introduction

Resumen. Los efectos del cambio climático han de ser estudiados a nivel global y local. Los modelos climáticos globales son capaces de realizar predicciones fiables a largo plazo de distintas variables termodinámicas, pero su gruesa resolución espacial dificulta su uso para estudios a escala regional. Para solventar este problema, se encuentran dos enfoques distintos de regionalización: dinámica, mediante el uso de modelos regionales climáticos, y estadística, la cual estudia las relaciones estadísticas entre las distintas variables. Gracias al auge de algoritmos de aprendizaje automático, se propone un método de inteligencia artificial generativa, novedoso en el campo, para la regionalización de la temperatura media diaria en las Islas Canarias.

I.1 Context

The human activities like burning fossil fuels and deforestation are closely related to the rise of greenhouse gases in the atmosphere [1], which causes climate change. It is then vital to get a better understanding of climate systems on both local and global scales. Despite local high resolution simulations are of significant importance in benefiting climate projection and analysis of extreme events, these carry a large amount of computational requirements and costs. Moreover, the results of many sectoral applications, such as hydrology, agriculture, energy or health, strongly rely on local climate variables with almost daily frequency.

Global Climate Models (GCMs) are numerical simulations, computed on a 3D grid, of the Earth's climate, based on well established physical principles, and have been demonstrated to reproduce observed features of recent and past climate changes, providing credible quantitative changes of some thermodynamic variables at large scale [2]. GCMs do not provide a representative description of local regions, due to the spatial resolution of the model, usually around 2 degrees both in latitude and longitude. Furthermore, the lack of representativeness of the simulation results is more evident in regions with complex terrain [3].

To address the coarse spatial resolution of GCMs, climate modelers have developed different approaches, among which two stand out: dynamical downscaling and empirical-statistical downscaling. Dynamical downscaling is based on the development of Regional Climate Model (RCM) simulations with high resolution, using meteorological variables extracted from GCMs as boundary conditions, resulting in convection-permitting models (CPMs) [4] when the spatial resolution is on the order of a few kilometers. Although this method offers a comprehensive representation of local climate conditions, requires substantial computational resources. In contrast, statistical downscaling (SD) finds empirical-statistical relations between regional climate variables (predictands) and the low resolution (LR) large-scale synoptic conditions (predictors). Using daily data, relationships are learned from observational or simulated values of predictors and high-resolution (HR)

predictands, allowing for the generation of locally downscaled values.

I.2 Deep Learning

The emergence of Deep Learning and Generative AI have brought to the table alternative approaches to SD, that have attracted climate physicists, due to its potential to identify possible non linear relations between climate variables, revealing previously unknown patterns. Along the years, various machine and deep learning methods have arise in different contexts such as computer vision and natural language processing, allowing a more efficient automatization of processes, and better performance on different tasks such as regression and classification. The application of these technologies to science is straightforward due to the amount of available data, and its vector nature, usually as *arrays* with shape (*height*, *width*, *channels*) in the case of images, or scalar maps.

The goal of machine learning consist of emulating the distribution of the data to be studied. A deeper structure learns more complex relationships, as there are more units that apply non linear activation functions to the data. The cycle of a machine learning algorithm has three distinct phases: training, validation and test. In the training, the aim is to fit the model parameters, also called weights, in order to reproduce the distribution of the data that goes through the model. At first, the parameters of the model are set random (although there exist multiple techniques, such as Xavier Initialization [5]), and each iteration optimizes the parameters to fit the data by minimizing a *loss*, a function that evaluates the quality of the model output by comparing it with the real target of the data. After the necessary amount of iterations of training, the validation stage analyzes the results through the selected metrics, depending of the *hyper parameters*, parameters of the model that stay fixed throughout the training process. Following to various training and validation processes, the user will choose the hyperparameters that made the model to perform better at validation. The model is now tested with a set of data that it has not seen before. If the results of the test are satisfactory, the model is ready for deployment.

Among the most used architectures of deep learning, the Convolutional Neural Networks (CNNs) have revolutionized all kind of tasks involving computer vision. CNNs consist on different layers (See Figure 1 as an example of usual CNN architecture):

- **Convolution:** Various filters (also called kernels) with smaller size of the input data, sweep through the input while computing an element-wise product with the area that the filter covers, then realize a sum over all the products, and the result is the output of the kernel at the point where it is centered. An iteration through all the possible pixels results on data with smaller size, depending on the *kernel size*, *stride* and the *padding*. The number of output channels is given by the number of filters applied.
- **Max Pool:** In this case, a kernel also goes through the image, and instead of the computation of the element-wise product, the kernel search the highest value in the area that the kernel covers, resulting in a smaller size output. There are similar versions that calculate the average of each block, as the Average Pool.

- **Flatten:** Once the data has crossed along various convolutions and max pools that lead to a final shape ($height'$, $width'$, $channels'$), the output is converted into a one dimensional vector with length $height' \times width' \times channels'$.
- **Fully Connected:** An usual *MultiLayer Perceptron* (MLP) network.

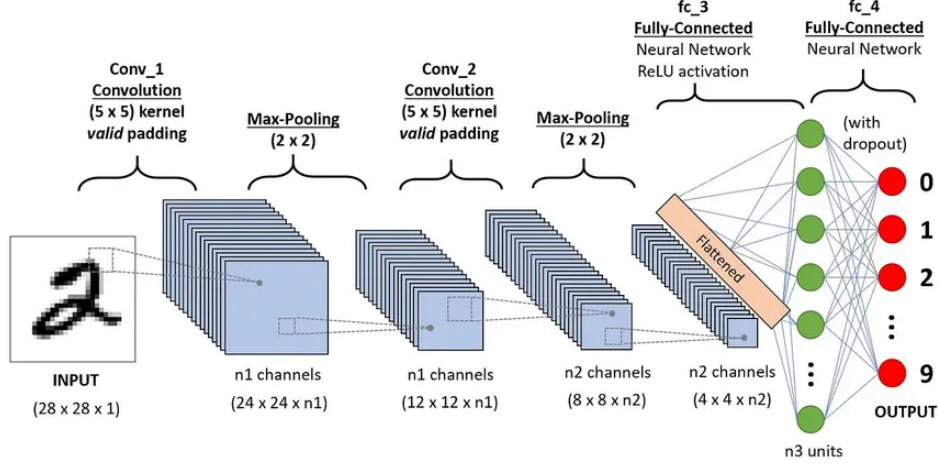


Figure 1: Example of a CNN architecture. Source: [6]

I.3 State of the art

The CNNs are able to outperform classic algorithms, such as multilinear and generalized regressions at VALUE metrics in the case of precipitation predictions, and reproduce extremes of temperature [7]. The natural evolution of the CNNs is the U-Net, a neural network architecture originally constructed to compute semantic segmentation tasks of biomedical images [8]. It consists on a encoder path that shrinks the resolution of the input image by blocks of 3×3 double convolutions, and 2×2 Max Pool layers, adding channels in the process, followed by a bottleneck with the same double convolution, and ends with a decoder path with the same number of blocks, but with upsamples instead of max pools for the image to match the corresponding block of the encoder, and finally match the input image resolution. To prevent losing spatial information of the image, the result of each block of the encoder is concatenated through a residual connection to the same block of the decoder. This model has been also applied to our area of study, as a RCM emulator, bringing promising results in computational benefit, reaching a 2 hour training stage, and less than a minute for one inference, yet problems with temperature extremes are notable [9].

On the other hand, generative AI takes advantage of the chaotic nature of the atmosphere by learning to synthesize statistically consistent possible weather patterns. Generative adversarial networks (GANs) provide plausible emulations of weather predictions, however, the generator produces a limited type of outputs, so it does not cover the full diversity and variety of the data distribution [10]. Denoising Diffusion Probabilistic Models (DDPMs)

conditioned on the LR predictors are a novel approach that is starting to be used to downscale a variety of thermodynamic variables of the atmosphere such as temperature, specific humidity, zonal and meridional winds, and precipitation, reducing costs and improving uncertainty estimates [11]. However, climate variables cannot be treated in the same way as channels in the image domain, although that's how they are used [12].

I.4 Goal

The Canary Islands have a small size in comparison with the resolution of GCMs, and a complex orography which complicates the predictions of regional values from the variables of interest. The *Grupo de Observación de la Tierra y la Atmósfera* (GOTA), from Universidad de La Laguna, is responsible of the dynamical downscaling of the Canary Islands using mesoscale weather models, which present the specified issues of dynamical models. The present work proposes a DDPM, to address the mentioned limitations, and predict 2 meter daily mean temperature, applied to the Canary Islands. The code of the model together with its evaluation and results can be found in the following GitHub repository:

https://github.com/samuelmzc/TFG_StatisticalDownscaling

II Methodology and Data

Resumen. En el presente apartado se expone la naturaleza y origen de los datos usados en el trabajo, junto al modelo implementado para solventar los problemas recurrentes de la regionalización climática estadística. A su vez, se muestran los procedimientos de entrenamiento e inferencia del modelo, su bajo coste y tiempo computacional. Las métricas usadas para la evaluación del mismo provienen de la iniciativa europea VALUE.

II.1 Evaluation metrics

The model is evaluated on the metrics of COST Action VALUE, an European Network to validate and develop different SD methods [13]. The aim of the VALUE validation framework is to encourage knowledge and collaboration between scientist of different disciplines in climate science on SD community. A subset of the metrics proposed are shown in table 1.

Description	Units
Bias (for the mean)	$^{\circ}C$
Bias (for the 2nd percentile, P2)	$^{\circ}C$
Bias (for the 98th percentile, P98)	$^{\circ}C$
Root Mean Squared Error (RMSE)	$^{\circ}C$
Ratio of standard deviations	–
Pearson Correlation	–

Table 1: Subset of VALUE metrics used in our study of temperature with their units (– refers to dimensionless metrics)

The bias estimates the systematic errors of the model and describes how far the predictions are from the real values on average, as it is computed for the mean of both the predictions and the WRF data. To disregard the negative values, the absolute value of the bias is used as the metric. For $\mathbf{y}^*$ the predictions of the model, and $\mathbf{y}$ the WRF simulations, the absolute value of the bias is defined as $Bias(\mathbf{y}^*) = |\bar{\mathbf{y}}^* - \bar{\mathbf{y}}|$, thus, the units are $^{\circ}C$. To study how the model predicts both the cold and hot extremes of daily mean temperature, the 2nd and 98th percentile bias have been computed also as the absolute difference.

The average magnitude of the error is given by the *Root Mean Squared Error* (RMSE). It is always a positive, and values close to zero indicate a good fit of the model. If N is the number of samples, is defined as

$$RMSE(\mathbf{y}^*) = \sqrt{\frac{1}{N} \sum_{i=0}^N (y_i^* - y_i)^2} \quad (1)$$

and was also taken into account as metric, with units of $^{\circ}C$. The ratio of standard deviations express the relative variability of two different probability distributions. Let σ_{y^*} and σ_y be the standard deviations of the model and WRF distributions respectively, then the ratio of standard deviations is written as it follows

$$Ratio(y^*, y) = \frac{\sigma_{y^*}}{\sigma_y} \quad (2)$$

A value larger than one leads to a higher variability of the predictions, and a smaller than one results to a higher variability of the WRF temperatures. At last, the correlation between the real and predicted distribution is calculated with the Pearson coefficient r . For $Cov(y^*, y)$ the covariance of both distributions, the Pearson coefficient has the form

$$r = \frac{Cov(y^*, y)}{\sigma_{y^*} \cdot \sigma_y} \quad (3)$$

and is a common indicator of the correlation between two series of data. A value close to 1 results in a high correlation, -1 a high negative correlation, and 0, non existent correlation.

II.2 Data

The VALUE framework suggests a total of 20 LR predictors, consisting on five thermodynamic variables (specific humidity, temperature, zonal and meridional wind, and geopotential height) at four different pressure levels (1000, 850, 700, and 500 mb), in a temporal span of almost 30 years. In our case, the recent past period 1980 – 2009 was selected. These conditions, like the HR data, are sampled daily. Therefore, there exists a LR synoptic map corresponding to each HR 2 m daily mean temperature map.

To evaluate SD methods reliability in changing climates, strategies like using extreme historical conditions, analyzing downscaled future trends, or using pseudo-observations have been proposed. Recently, hybrid approaches like RCM (regional climate model) emulators have emerged, combining the strengths of dynamical and statistical methods [9]. These emulators, trained on RCM outputs instead of observations, as is the case of the perfect prognosis approach, overcome some limitations of SD and allow cost-efficient ensemble generation. In this work, climate regionalizations performed by the WRF (Weather Research & Forecasting) mesoscale model were used as pseudo-observations, so that the trained model emulates its behavior. Specifically, the results of a WRF simulation with a spatial resolution of 3 km in the innermost domain and using MIROC-ESM [14] as the driven global climate model [15] were used.

Thus, the inputs to the model, LR, are the daily means of the synoptic variables mentioned above at the four pressure levels, obtained directly from the r1i1p1 realization of the MIROC-ESM global climate model, belonging to CMIP5 (Coupled Model Intercomparison Project Phase 5). The domain ranges from 22N to 40N in latitude and from 31W to 1E in longitude, displayed in Figure 3. The original resolution of the data is

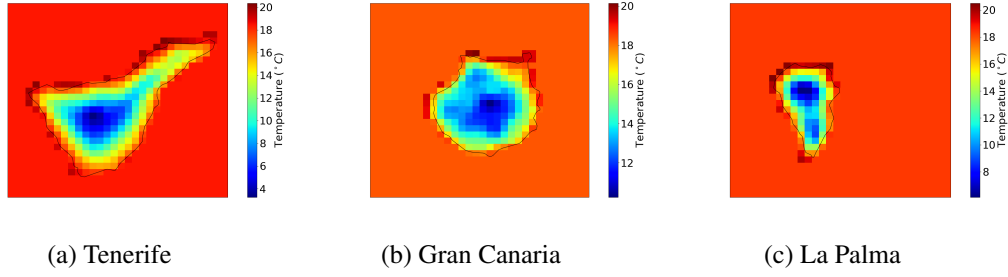


Figure 2: WRF simulations of daily mean temperature of a particular date.

$2.8^\circ \times 2.8^\circ$ although they were resampled using a bilinear interpolation to a resolution of $0.58^\circ \times 1^\circ$ to obtain a 32×32 grid points to match the size of the output maps, HR, for reasons to be discussed later.

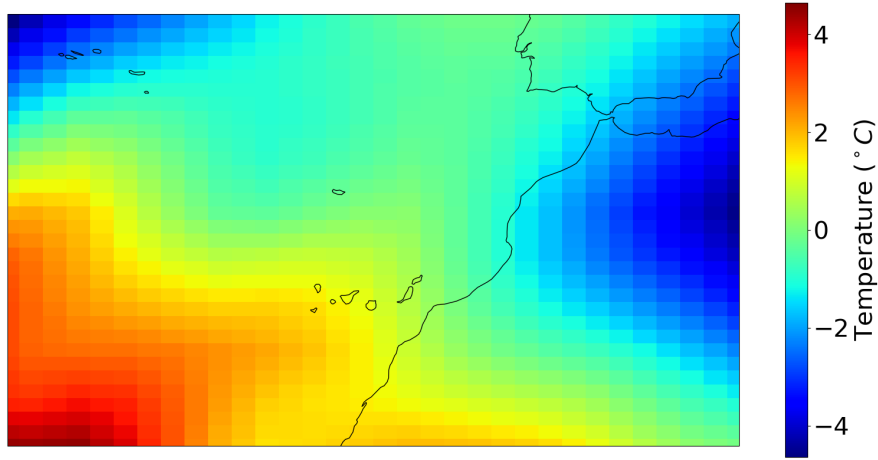


Figure 3: Region of the LR data along with the temperature at 500 *mb*

The model outputs correspond to the daily mean temperature maps directly at the resolution provided by the WRF simulations, 3×3 km. Three regions of 32×32 grid points have been taken, one encompassing the island of Tenerife, another Gran Canaria and the last one comprising the island of La Palma (Figure 2).

II.3 Diffusion model

The *Denoising Diffusion Probabilistic Model* (DDPM) is a probabilistic model designed to allow extreme flexibility in model structure, and exact sampling. A Markov chain is able to convert any given probability distribution into a totally different distribution, an idea whose origin arises from non-equilibrium statistical physics.

Applied to our case study, the Markov chain will convert a standard normal distribution, into a desired and unknown distribution, that will hopefully fit to the data distribution of the HR data $\mathbf{x}^{HR}$. At first, a forward process is made in a certain number of time steps T , that converts the original data distribution $q(\mathbf{x}_0)$, into a Gaussian distribution $q(\mathbf{x}_T) \sim \mathcal{N}(0, \mathbb{I})$. Each time step gradually adds gaussian noise to the previous step via a fixed variance scheduler β_t , and has the form $q(\mathbf{x}_t^{HR}|\mathbf{x}_{t-1}^{HR}) = \mathcal{N}(\mathbf{x}_t^{HR}; \sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1}^{HR}, \beta_t \mathbb{I})$ for $t = 1, 2, \dots, T$. The concatenation of all the time steps gives a Markov chain:

$$q(\mathbf{x}_T^{HR}) = q(\mathbf{x}_0^{HR}) \prod_{t=1}^T q(\mathbf{x}_t^{HR}|\mathbf{x}_{t-1}^{HR}) \quad (4)$$

Due to the nature of the Markov chain, it is possible, and also, more convenient, to write the forward process to any time step t from the original data $\mathbf{x}_0$ in closed form, introducing $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=0}^t \alpha_i$, to finally get $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0, (1 - \bar{\alpha}_t) \cdot \mathbb{I})$. In this way, it is possible to reparametrize the calculation as

$$\mathbf{x}_t^{HR}(\mathbf{x}_0^{HR}, t) = \sqrt{\bar{\alpha}_t} \mathbf{x}_0^{HR} + \sqrt{1 - \bar{\alpha}_t} \epsilon \quad (5)$$

where $\epsilon \sim \mathcal{N}(0, \mathbb{I})$. This reparametrization does indeed simplify the notation, and introduces a proper way to obtain the result of applying t time steps of forward Gaussian diffusion to the original data.

A linear noise scheduler works well for high resolutions, while it is suboptimal for resolutions of 64×64 and 32×32 , i.e., our case, as the forward diffusion is too noisy, and does not contribute to the sampling. To address this problem, the following noise scheduler has been proposed [16]:

$$\bar{\alpha}_t = \frac{f(t)}{f(0)} \quad f(t) = \cos\left(\frac{t/T + s}{1 + s} \cdot \frac{\pi}{2}\right)^2 \quad (6)$$

where $s = 0.008$. This is a cosine scheduler, designed to have a linear drop off of $\bar{\alpha}_t$ in the middle of the diffusion, but changing very little at $t = 0$ and $t = T$. Note that $\beta_t = 1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}$.

The probabilistic model $p_\theta(\mathbf{x}_0^{HR}|\mathbf{x}^{LR})$, with θ the parameters of the model, consist on the reverse process of the forward diffusion starting at $p_\theta(\mathbf{x}_T)$, where a backbone model has learned the mean and covariance of each time step, conditioned by the LR data $\mathbf{x}^{LR}$:

$$p_\theta(\mathbf{x}_0^{HR}|\mathbf{x}^{LR}) = p_\theta(\mathbf{x}_T^{HR}) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}^{HR}|\mathbf{x}_t^{HR}, \mathbf{x}^{LR}) \quad (7)$$

$$p_\theta(\mathbf{x}_{t-1}^{HR}|\mathbf{x}_t^{HR}, \mathbf{x}^{LR}) = \mathcal{N}(\mathbf{x}_{t-1}^{HR}; \boldsymbol{\mu}_\theta(\mathbf{x}_t^{HR}, \mathbf{x}^{LR}, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t^{HR}, \mathbf{x}^{LR}, t)) \quad (8)$$

At first, the covariance matrix will be fixed as $\boldsymbol{\Sigma}_\theta(\mathbf{x}_t^{HR}, \mathbf{x}^{LR}, t) = \sigma_t^2 \mathbb{I}$, with $\sigma_t^2 = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t} \beta_t$, which, as we assumed that β_t is constant at each time step t , it is an untrained parameter. Then, an initial proposal for the loss function of the backbone network to minimize involves the *Mean Squared Error* (MSE) between the predicted mean $\boldsymbol{\mu}_\theta(\mathbf{x}_t^{HR}, \mathbf{x}^{LR}, t)$ and the original mean $\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t^{HR}, \mathbf{x}^{LR}, \mathbf{x}_0)$ on the following form:

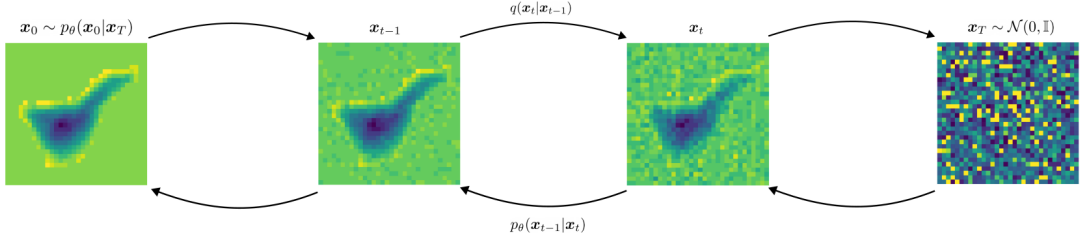


Figure 4: Demonstration of the forward and backward process of the diffusion with a particular temperature map of Tenerife.

$$L(\theta) \sim \|\tilde{\mu}_t(\mathbf{x}_t^{HR}, \mathbf{x}^{LR}, t) - \mu_\theta(\mathbf{x}_t^{HR}, \mathbf{x}^{LR}, t)\|^2 \quad (9)$$

The original mean will explicitly depend on the added noise ϵ at the time step due to 5, leading to the conclusion that the predicted mean will try to predict the noise added at that same time step $\epsilon_\theta(\mathbf{x}_t^{HR}, \mathbf{x}^{LR}, t)$, leading to the following loss function.

$$L(\theta) \sim \|\epsilon - \epsilon_\theta(\mathbf{x}_t^{HR}, \mathbf{x}^{LR}, t)\|^2 \quad (10)$$

Once the backbone model is trained, it will reproduce the predicted noise $\epsilon_\theta(\mathbf{x}_t^{HR}, \mathbf{x}^{LR}, t)$.

The sampling process consist on generating a random matrix, with the role of the original data after T time steps, i.e, $\mathbf{x}_T^{HR} \sim \mathcal{N}(0, \mathbb{I})$. Then, the following step of the backward process is given by the predicted mean of the model and σ_t in the following form:

$$\mathbf{x}_{t-1}^{HR} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t^{HR} - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t^{HR}, \mathbf{x}^{LR}, t) \right) + \sigma_t \mathbf{z} \quad (11)$$

with the calculation of $\mathbf{x}_{t-1}^{HR}$ for $t = T, T - 1, \dots, 1$, where we will set $\mathbf{z} = 0$ for the last time step to sample. This algorithm will return the predicted value $\mathbf{x}_0^{HR} \sim p_\theta(\mathbf{x}^{HR}|\mathbf{x}_T^{HR}, \mathbf{x}^{LR})$. The total cycle of the model is graphically summarized in Figure 4.

As backbone of the diffusion model, it has been used an U-Net. The architecture used varies from the original as our model has to meet some requirements to work properly. First, the U-Net has to know at which denoising time step it encounters itself. For this purpose, a time-positional embedding has been added, based on the Positional Encoding of the Transformer architecture [17]. Each time step t has associated a vector of dimension d_{emb} , called the embedding dimension, where the components are sines or cosines, depending if the index of the component is even or odd, of different frequencies. It has the form:

$$PE_{t,2i} = \sin\left(\frac{t}{10000^{2i/d_{emb}}}\right)$$

$$PE_{t,2i+1} = \cos\left(\frac{t}{10000^{2i/d_{emb}}}\right)$$

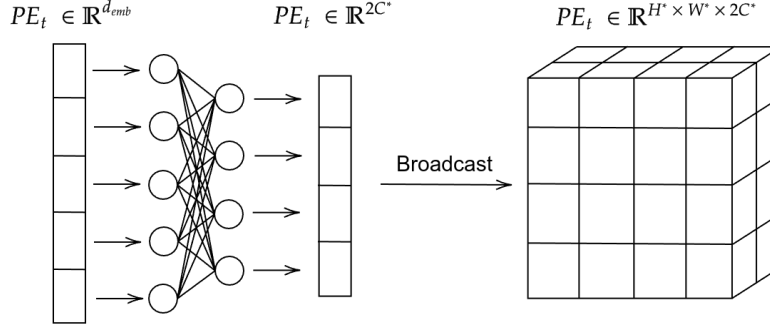


Figure 5: Operations computed to the time-positional embedding vector of timestep t .

where $i = 1, 2, \dots, d_{emb}/2$. The vector is then passed through a dense layer, leading to a vector of size $2C^*$, and then broadcasted to compute operations with the tensor with shape (H^*, W^*, C^*) (Figure 5), result of a convolution applied to the input data. . The reason of the vector having size $2C^*$ is because it will be split in half; one half will be added to the input as a shift term, and the other multiplied as a scale term.

The fundamental unit of the model is a block constructed with: convolutions with 3×3 kernel, a *Sigmoid-Weighted Linear Unit* (SiLU), which is an activation function applied to the result of the convolution, and a batch normalization in order to fix the distribution of the incoming data. The time-positional embedding is added with the scale and shift terms, and the process repeats one more time. The block has a residual connection that passes the input through a different convolution, and its added to the block output, so the training is more efficient as the backwards propagation can move on these connection [18]. This is a *Residual Network* (ResNet), thus the block is called a ResNet block (see Figure 6).

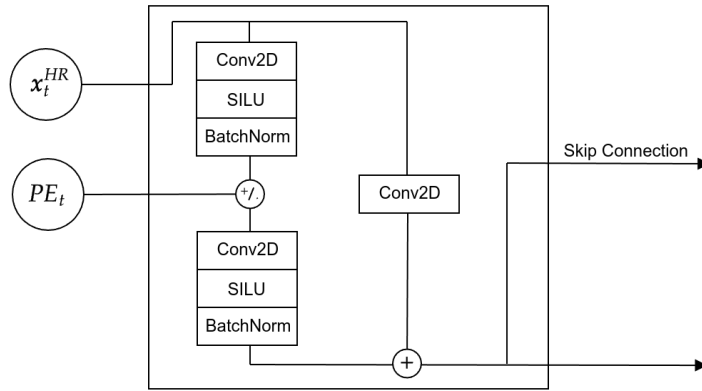


Figure 6: Diagram of the ResNet Block.

At the input of the U-Net, the synoptic conditions, described as the LR data, are simply concatenated to the HR map in the channel dimension, resulting in a $32 \times 32 \times 21$ array. Then, the U-Net only sees the conditions at the entry. The encoder of the model consist of blocks, each of these followed by a 2×2 max pool, starting at a 32 convolutions block, to a 1024 convolution block, that outputs a $2 \times 2 \times 1024$ array. The bottleneck also apply a 1024 convolution block. The decoder has the same number of blocks, but max pools are replaced by up-sample layers, and the result of the corresponding block of the encoder gets concatenated in the channel axis with the previous up sample. The last part of the model is a *same* convolution, to leave unchanged the resolution of the image, and adjust the number of channels of the output, corresponding to the number of images to predict, in our case, 1, the noise added at time step t . The U-Net of the model is represented at Figure 7. The number of parameters obtained for the model is 52.533.121.

To provide significant relation between the different axes of the input, a visual attention layer was implemented to the model, named *Triplet Attention* [19], as it splits the input data into three different sections, each of these computing a convolution and a permutation. The results were the same as if the attention was not included, so it was left behind.

II.4 Training and sampling

The training consist on the backbone model learning to predict the noise of certain time step t , as described in the previous section. The original data set was split into 80/20 train and test sets. Thus, then the training set consist of 8766 maps between 1980 and 2004, and a test set of 2192 from 2004 to 2009. The validation process has been done with the test set. The hyperparameters were chosen by grid search in the hyperparameter space, leading to choose the following values:

- **Batch size:** 128
- **Epochs:** 50
- **Total time steps:** $T = 1000$
- **Learning rate :** $\alpha = 5 \cdot 10^{-4}$ (fixed)
- **Embedding dimension:** $d_{emb} = 2048$

For each batch of data, the forward diffusion process was obtained, resulting in the noise added, and the noised output. The time step used is a random vector with same size as the batch, where the values where bounded between 0 and T , this means that each data element of the batch was driven to a different and random time step than the other, instead of training for all the time steps, that could lead to an exponential time growth of the computation time. The optimizer of the parameters used is *Adaptive moment estimation* (Adam), widely used in machine learning due to its velocity and efficiency, as it uses weighted averages, instead of just the derivative of the lost function as *Gradient Descent* does [20]. The training

process can be seen at Algorithm 1

Algorithm 1 Training

- 1: $\mathbf{x}_0^{HR} \sim q(\mathbf{x}_0^{HR})$
 - 2: $t \sim \text{Random}(\text{batch size}, 0, T)$
 - 3: $\epsilon \sim \mathcal{N}(0, \mathbb{I})$
 - 4: Take Adam step on

$$\nabla \|\epsilon - \epsilon_\theta(\mathbf{x}_t^{HR}, \mathbf{x}^{LR}, t)\|^2$$
 - 5: **repeat until convergence**
-

Algorithm 2 Sampling

- 1: $\mathbf{x}_T^{HR} \sim \mathcal{N}(0, \mathbb{I})$
 - 2: **for** $t = T, \dots, 1$ **do**
 - 3: $\mathbf{z} \sim \mathcal{N}(0, \mathbb{I})$ if $t \neq 1$ else $\mathbf{z} = 0$
 - 4: $\mathbf{x}_{t-1}^{HR} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t^{HR} - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(\mathbf{x}_t^{HR}, \mathbf{x}^{LR}, t) \right) + \sigma_t \mathbf{z}$
 - 5: **return** $\mathbf{x}_0^{HR}$
-

The model will also be conditioned on the island. A common approach to these type of problems is the use of embeddings for the model to figure out the different conditions, in our case, the islands of Tenerife, Gran Canaria and La Palma. For our case, it's enough to train the model on each island, and save the weights with the corresponding name, instead of only doing one training with an embedding, as the computation time for each training is manageable. A full training for each island of 50 epochs at a computer with 32 cores Intel Xeon CPU E5-2683 v4 - 2.10GHz takes an hour to finalize.

The sampling method, as mentioned, and written at Algorithm 2, takes between 4 and 8 seconds to infer a single sample. The inference of the test set of data takes 4 hours. This time mostly depends, by definition, on the total number of denoising time steps, and also, on the dimension of the time-positional embedding vector. At the validation step, a test was performed to study the time required to sample the entire test set choosing as hyperparameters $T = 100$ and $d_{emb} = 256$, which resulted about 40 minutes, but bringing worse qualitative and quantitative results. Each inference of the test set is saved as a channel of an array, leading to a total shape of $32 \times 32 \times 2192$. This method improves the computational time, as Python works fast with arrays, and no more inferences have to be done.

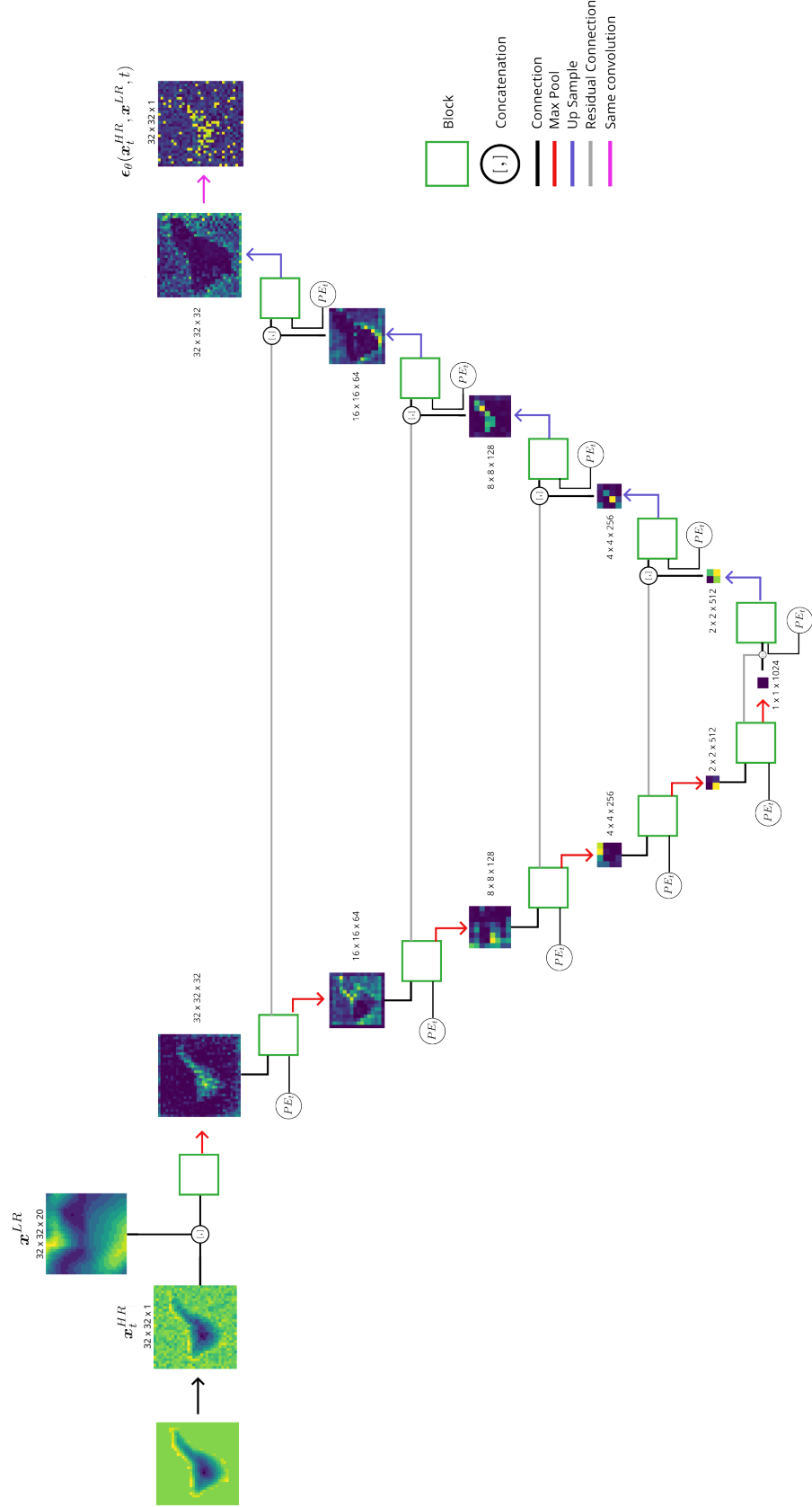


Figure 7: Scheme of all the process to predict the noise, from the initial data, to the noise prediction.

III Results

Resumen. Se presentan los resultados del modelo, y se procede con análisis cualitativos y cuantitativos de estos. Para ello, se evalúan cualitativamente dos inferencias del modelo, una más precisa y otra menos, para estudiar los puntos fuertes y débiles que a priori podría tener el modelo. Para el estudio cuantitativo, se estudian los resultados de las métricas propuestas en la sección anterior, y para examinar la reproducibilidad del modelo, se realiza un gran número de inferencias para las mismas condiciones sinópticas, de forma que la media de las temperaturas debería ser cercana a la temperatura media diaria predicha por WRF.

This section presents the results of the model, its ability to reproduce and predict HR temperature maps on short times compared to dynamical models. The metrics are calculated using the test set, with a one-time inference, although the mean of various inferences for each map was also taken into account as a test, but the results do not vary significantly. The results for the three islands are always presented in the same corresponding map, despite the predictions and the training were performed individually.

The model predicts accurately the simulations from WRF. From a qualitative point of view, the diffusion is able to learn in an acceptable manner the complicated orography of the islands. It is also able to differentiate between the climate of different regions of each island. In Tenerife, the lowest temperatures are located at *Parque Nacional del Teide*, a nature protected area with a mean altitude of 2000 meters above sea level, and a maximum of 3718 meters at the summit of El Teide, responsible of the low temperatures. The same is true for La Palma, where the lowest temperatures happen at *Parque nacional de la Caldera de Taburiente*, another nature protected area with a mean altitude of 750 meters, and a peak at 2428 meters at El Roque de los Muchachos. The island of Gran Canaria has a lower altitude average than the other two islands, and the highest point is located at *Pico de las Nieves*, with 1957 meters of altitude, therefore with higher minimum temperatures.

Figure 8 presents two different cases, with their corresponding WRF simulation, and model inference. The first situation belongs to the month of March, close to spring's equinox, where temperatures are relatively low. The inference in this case is very accurate, mainly in Tenerife, and predicts lower temperatures in Gran Canaria, and La Palma, at Roque de los Muchachos. The model tends to underestimate the temperature specially in these last two islands, despite of using exactly the same model configuration.

The second case occurs in July, in the middle of summer, a period of higher temperatures. At first sight, it is safe to say that the model underestimates the temperature of all the islands. This possibly corresponds to an extreme of mean daily temperature in summer due to a heat wave, thus inference returns much lower temperatures. This behavior occurs systematically in all the test set in summer.

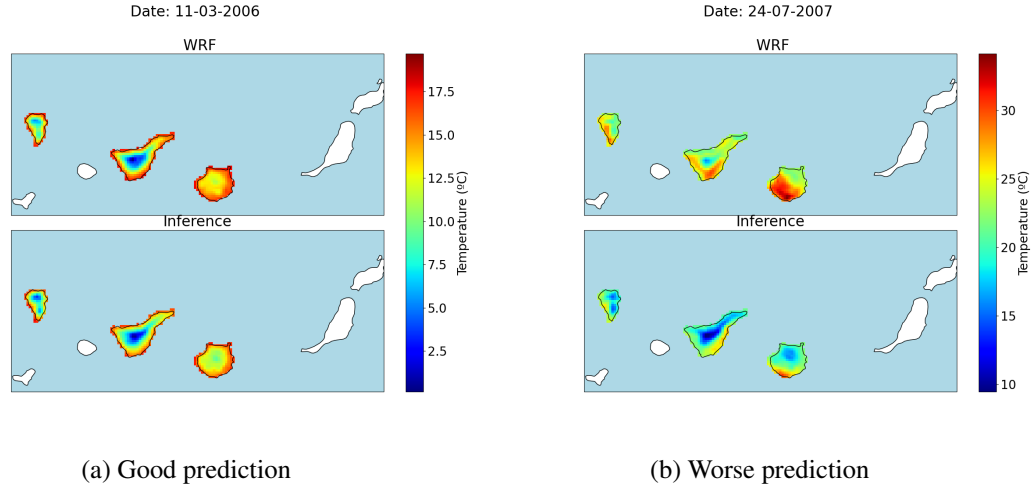


Figure 8: Predictions of the model with corresponding WRF simulations of the same day to perform a quantitative analysis. The figure on the left is a qualitative good inference (March), and at the right, a worse prediction (July).

III.1 Performance on VALUE metrics

The VALUE metrics obtained for each island is the mean of the result of the metric for every pixel (Table 2). The bias of the inference with respect to WRF simulation is around half a Celsius degree. The larger bias is obtained in Tenerife, while the lower corresponds to Gran Canaria, with La Palma in between. In Tenerife, trade winds blows from the north east, leading to the formation of stratocumulus clouds between 500 and 1500 meters of altitude, due to the push they exert to the mountain sides of the north. This physical phenomenon whose effects on local temperature may be more difficult to infer from synoptic conditions, could also be responsible of the higher bias. This can be seen in Figure 9, where the highest bias values are observed in the mid-altitude areas of the north-facing slopes of the island of Tenerife. The figure also shows a larger bias at El Roque de Los Muchachos and, in general, in the highest areas of La Palma. In Gran Canaria, the maximum bias is lower, as expected from the mean value, and the lower bias zone that forms a ring around the middle of the island, with some shores with higher bias, similar to the observed at the center of the island. The uncertainty of the averaged values for all points of the grid corresponding to each island, expressed as twice the standard deviation, i.e, a 95% confidence interval, is greater than the mean in all the bias values. This is expected as the islands have a large variability in temperature and also the biases are highly variable (Figure 9).

The mean bias of the predicted 2nd percentile of temperature (Figure 10) is lower in the case of Tenerife and Gran Canaria. As seen in the qualitative analysis, the model performs better on low temperatures, rather than hot extremes. On the other hand, the mean of the 98th percentile is higher on every island. As the bias for the 2nd percentile is lower than the 98th percentile, it is concluded that the model is not predicting correctly hot extremes of daily mean temperature. In the case of La Palma, the daily mean values of temperature are lower on average, decreasing the number of hot extremes samples, and therefore the model performs better, leading to more accurate predictions and presenting a lower bias at 98th

Metric	Tenerife	Gran Canaria	La Palma	Units
Bias	0.61 ± 0.75	0.44 ± 0.46	0.52 ± 0.64	$^{\circ}C$
2nd Percentile bias	0.42 ± 0.61	0.34 ± 0.55	0.75 ± 0.56	$^{\circ}C$
98th Percentile Bias	1.02 ± 1.22	1.14 ± 2.03	0.78 ± 1.08	$^{\circ}C$
Root Mean Squared Error	2.82 ± 1.15	3.15 ± 1.16	2.96 ± 0.97	$^{\circ}C$
Ratio of standard deviations	0.98 ± 0.08	1.14 ± 0.21	1.07 ± 0.16	—
Pearson coefficient	0.79 ± 0.09	0.78 ± 0.08	0.77 ± 0.08	—

Table 2: Mean VALUE metrics for each island given as mean $\pm$ 2 · standard deviation.

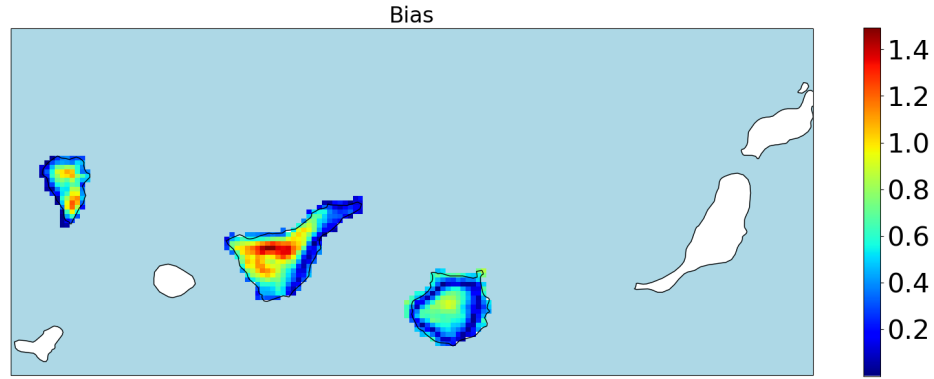


Figure 9: Absolute bias, in Celsius degrees

percentile than Tenerife and Gran Canaria (Figure 11).

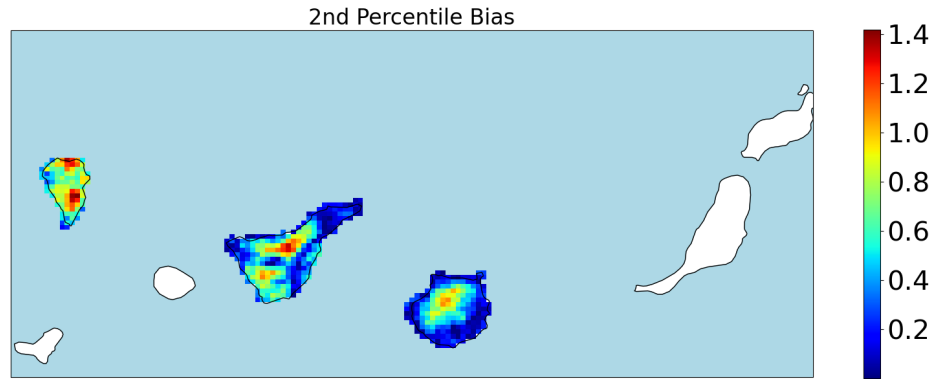


Figure 10: Absolute bias of the 2nd percentile, in Celsius degrees

The RMSE does not have the same variability as the biases (Figure 12), as the average magnitude of the errors made by the model does not vary throughout the islands. The mean values are relatively close, taking in consideration the standard deviations, which indicates

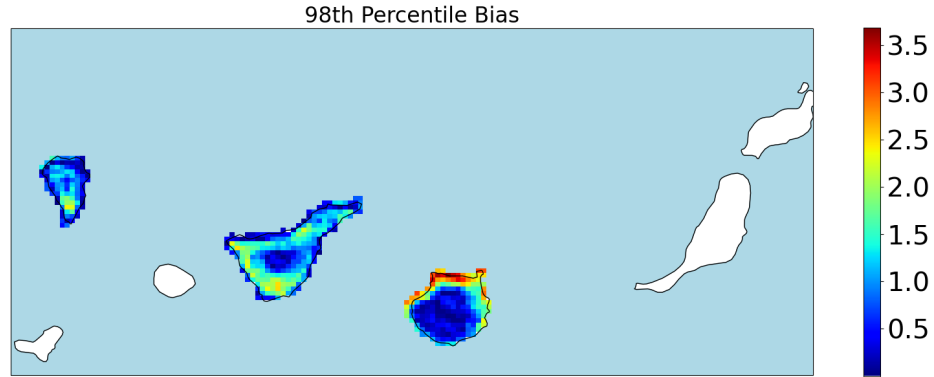


Figure 11: Absolute bias of the 98th percentile, in Celsius degrees

that the model predictions diverges from the WRF simulations on the same magnitude for all the three islands. In Gran Canaria, there is a peak of 4°C at Parque Natural Tamadaba, and descends to the east of the island until the shore, with a minimum of 2°C . Tenerife and La Palma keep the RMSE almost constant at 3°C for the vast majority of the islands, and the same minimum found at Gran Canaria, near the coasts. The Mean Squared Error (MSE) can be written as

$$MSE = Var + Bias^2$$

with Var the variance. The RMSE should always be higher than the model bias, which explains these high RMSE values.

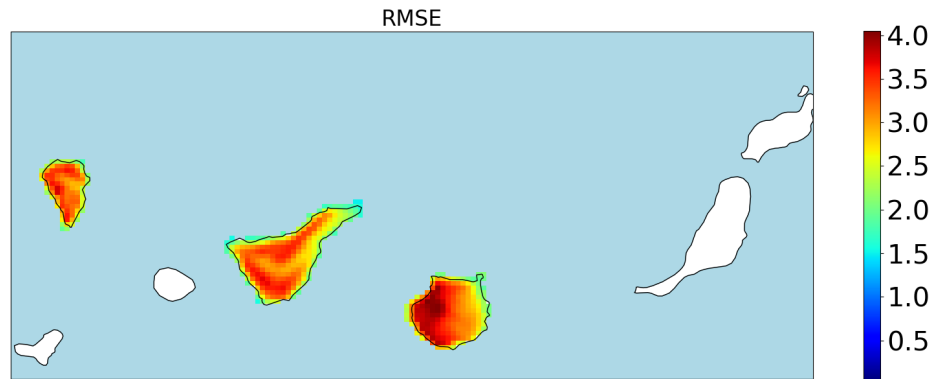


Figure 12: Root Mean Squared Error, in Celsius degrees

The ratio of standard deviations is close to one in most cases. A close value to one indicates that the predicted distribution has a similar standard deviation to the distribution of the WRF. Values of the ratio below 1 represent a standard deviation of the predictions

smaller than the WRF's standard deviation, and a value larger than 1, a larger predicted standard deviation. In all the cases, the uncertainty not allows one to conclude if the predicted standard deviation is smaller or larger than one, so a further analysis of all the pixels of each island must be done. Tenerife has the lowest uncertainty on the mean, and as it can be seen in Figure (13), most of the pixels of the island are slightly below one, except for some points at shores, and at El Teide. The opposite happens at Gran Canaria, as most of the surface results in a ratio value larger than one, result that reflects in the mean of the island. The highest values are found in the north of the island, indicating that the variance of the diffusion model data is about 20–30% higher than the variance of the data simulated by WRF. It also has the largest uncertainty due to the high variability of ratios. In La Palma the ratios show a behavior in between Tenerife and Gran Canaria, as there is also a high variability of values, but with more pixels at values smaller than one.

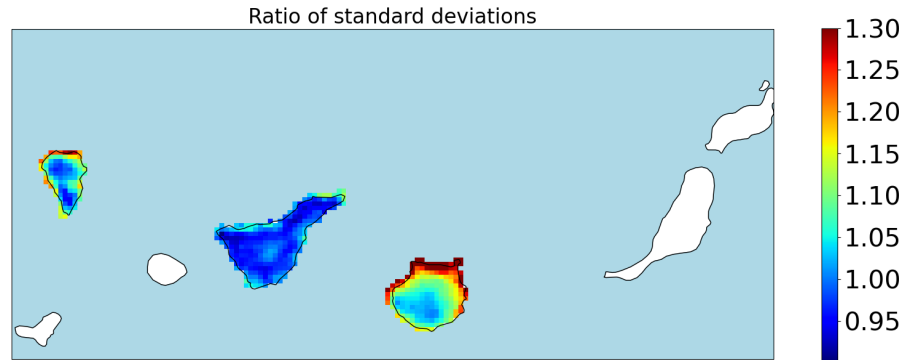


Figure 13: Ratio of standard deviations

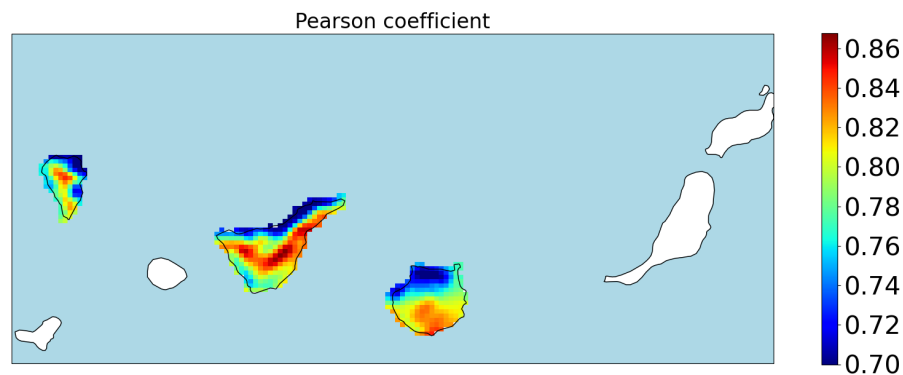


Figure 14: Pearson coefficient

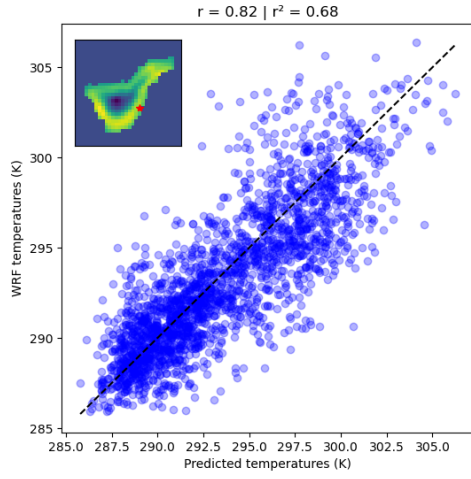
The Pearson coefficient is larger than 0.7 for every pixel of all the islands. The mean values are close to 0.8, asserting an acceptable correlation between the predicted and the

WRF daily values. The uncertainty is low, as the coefficient on each island varies in the order of 0.16 at maximum. Tenerife presents maximums of 0.86 at the central latitudes of the island. The worst correlations are concentrated in the north of the island, probably due to the stratocumulus clouds mentioned in the qualitative results. In La Palma, best correlation values appear at Roque de Los Muchachos, and worse, but still acceptable values at north coasts. Gran Canaria is divided into two, the north presents worse correlation than the south.

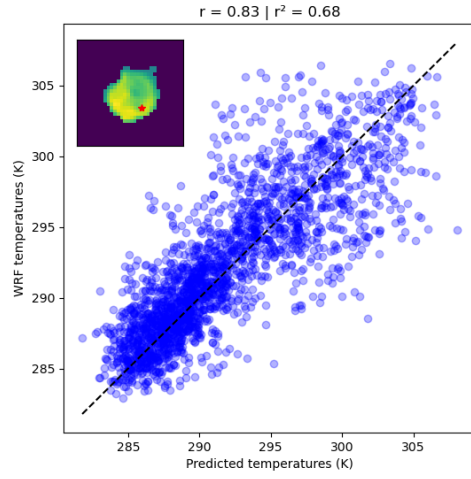
The correlations at points with high Pearson coefficient values on each island were studied (Figure 15). Each blue dot represents the predicted temperature for the WRF daily mean temperature simulation at a certain time, then the number of points is equal to the length of the test set, 2192. A dashed line representing the points where the predictions and the WRF temperature values are the same is plotted. Except in Tenerife, where the Pearson coefficient is 0.82, the majority of the blue dots are above the dashed lines, this means that for a given WRF temperature, the model predicts more values of daily mean temperatures under the WRF simulation, than higher values. These results continues to assert the underestimation of the model for temperatures.

III.2 Distribution of a single sample. Case study

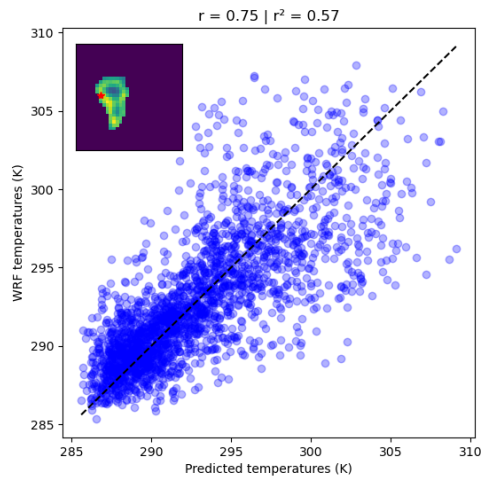
As explained above, the proposed model calculates inferences from a matrix of random values, plus information from synoptic conditions, reducing noise in successive steps. Then, it is necessary to study the distribution of the model output, mean daily temperature, for various inferences when the LR synoptic conditions stay fixed, i.e., the distribution of the model when it infers the same sample, for the same day, multiple times (starting from different random matrices). The computation of a thousand inferences for the same sample of WRF simulation, takes 4 hours for each island. As in the sampling, all the inferences were stored as channels of an array with total shape $32 \times 32 \times 1000$, which facilitates the necessary manipulations of the data. A histogram of 50 bins was computed for every selected point (Figure 16), using the same locations as in Figure 15. The mean value of the 1000 inferences of daily mean temperature, and the WRF simulation value are represented as red and black vertical straight lines, respectively. It is observed that, for a large amount of inferences, the model follows a normal distribution, with the mean close to the WRF simulation daily mean, and a standard deviation of approximately $0.7^\circ C$ in all the cases. The WRF temperatures of the histograms are around 292 K , an average daily mean temperature, where the model performs better. The closest distribution mean corresponds to Tenerife, with a deviation with respect to the real value less than a Celsius degree. Gran Canaria and La Palma get worse results, with a deviation with respect to the WRF simulation of about 2 K .



(a)

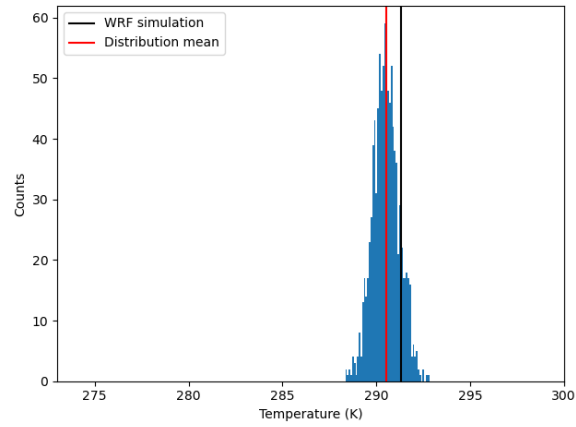


(b)

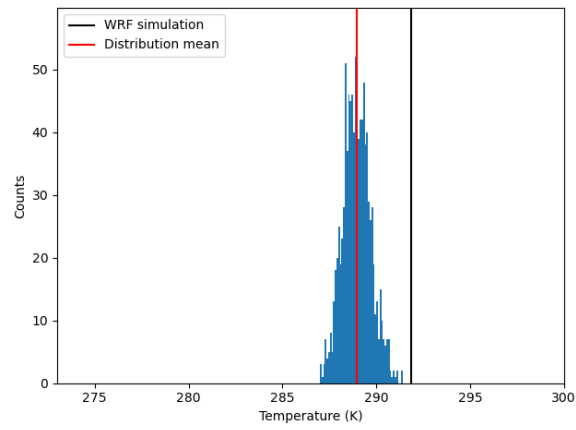


(c)

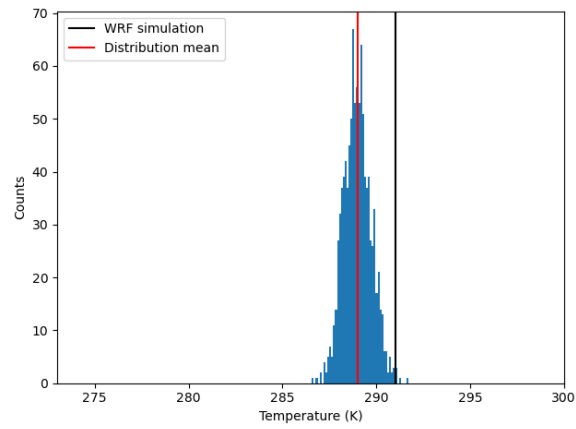
Figure 15: WRF temperature against the predicted temperatures of the model for each island, at the point marked as a red star in the corresponding map. Also the Pearson coefficient r , and r^2 are shown. The blue dots correspond to the temperatures along time for each pixel, and the dashed line indicates $WRF = Prediction$.



(a) Tenerife



(b) Gran Canaria



(c) La Palma

Figure 16: Histograms of 1000 samples of the same day for the same points in Tenerife, Gran Canaria, and La Palma as figure 15. The red line represents the mean of the 1000 points distribution, and the black line, the value of the temperature provided by WRF simulation.

IV Conclusion

Resumen. En esta última sección se muestran las conclusiones del trabajo, reflejando el motivo de este, la metodología, y los resultados obtenidos, con propuestas de mejora para futuros trabajos.

The present work explores the applicability of a conditional DDPM (Denoising Diffusion Probabilistic Model), a novel technique in climate physics, and generative AI, to address the limitations of previous climate SD techniques. The model is able to infer downscaled daily mean temperatures in Canary Islands, more precisely, in Tenerife, Gran Canaria, and La Palma, complex regions due to the size and orography of the archipelago, in a reduced time of computation. The training learns the diffusion noise added to WRF simulations with 3 km of spatial resolution, at all the $T = 1000$ time steps and takes approximately one hour for each island. The sampling of a single inference conditioned on the LR synoptic conditions of a particular day takes between 4 and 8 seconds for every island, a low computational cost compared to dynamical downscaling. The performance was evaluated using the VALUE metrics for SD, showing promising results of the proposed method.

It should be noted that the model proposed in this work is simple. Despite of the large amount of parameters of the model, the architecture is the one of a classical U-Net, adding the time-positional embedding, and only showing the conditions at the input. Better performance on the metrics could be obtained adding the conditions at every block, adapting the resolution of the data at each of them, for example, with a convolution. Attention modules could also been added, such as attention gates [21], or a general implementation of self attention to 2D data, able to find more intrinsic and deep relations between the different climate variables, and themselves, i.e., study how the variability of a certain variable affects to itself.

The qualitative and quantitative results show the robustness of the model for average and low temperatures. On the other hand, the model generate underestimated values of the daily mean temperature at extremes values, presumably in heat waves, dragging down the inconveniences of other techniques. This leads to the model not having a full range of the possible climate fluctuations. More precisely, the metrics where the model perform better are the ratio of standard deviations, and the Pearson coefficient. Both of these metrics are related with the correlation of the model with the WRF simulations, which indicates that despite of the worse results that the model infers for extreme hot temperatures, it is learning the climate variability along the different seasons of the year. The model also exhibits reproducibility for single samples, as the mean of all the inferred temperature maps is close to the WRF daily mean temperature of the same LR conditions for a large amount of inferences.

Despite the worse performance on extreme daily mean temperatures, the low computational costs and times make the model an interesting approach to study the evolution of thermodynamic variables of the atmosphere at the long term, leading to direct and indirect effects of green house gases, and climate change.

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